

Interception Model of Random Scanning Strategies Against Frequency-Agile Radar in Electronic Support

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In the context of electronic support, the primary objective is to extract as much information as possible from the radio frequency spectrum. Passive receivers are employed to monitor a wide frequency range that includes both communication and radar signals. Among the available receiver technologies, superheterodyne receivers (SHRs) are notable for their low cost and long-range capabilities compared to conventional wide-band receivers. However, SHRs have a narrow instantaneous frequency band, which needs to be scanned across the spectrum of interest. Their main drawback is their low probability of interception. This highlights the need for predictive models that can estimate interception performance against known emitters, to improve the optimization of receiver parameters. Existing models consider either deterministic scanning strategies (Hatcher, 1976), (Self and Smith, 1985), (Clarkson et al., 1996), or random scanning with simple emitters (El-Mahassni et al., 2004). In this work, we propose a model that accounts for randomly scanning multichannel SHRs, targeting frequency-agile radar emitters with rotating directional antennas. To do so, we use mathematical results on the coincidence between two pulse trains with a single window. We provide insights on the influence of main parameters and a comparison with existing models. Model validation is carried out in a simulated electronic warfare environment. We apply a minimax scheme to optimize a random scanning strategy for the interception of multiple known emitters, with uncertainty intervals. The proposed model allows to consider complex

Received 23 June 2025; revised 24 October 2025; accepted 25 January 2026. Date of publication 3 February 2026; date of current version 7 April 2026.

DOI. No. 10.1109/TAES.2026.3660987

Refereeing of this contribution was handled by J. Xu.

This work was supported by the DGA/AID (Directorate General of Armaments / Defense Innovation Agency).

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interception scenarios, and may serve as a theoretical baseline and design tool for adaptive scanning strategies.

I. INTRODUCTION

Electronic Warfare (EW) is a critical part of modern military operations. Its primary objective is to dominate the electromagnetic spectrum using various techniques, including radar systems, jamming technologies, or stealth capabilities. One branch of EW is electronic support (ES), which involves the detection, location, and recognition of electromagnetic emissions from radars and radio communications using passive receivers. Radar systems are commonly deployed on aerial, naval, and ground platforms, while communication signals come from TV broadcasting, GSM, 4G, 5G, and WiFi devices. These various signals are emitted in a wide frequency range, from approximately 3 kHz to 100 GHz.

In this context, receivers capable of covering wide frequency bands are essential for collecting the maximum amount of information. Wideband receivers are well-suited to this requirement, as they can monitor bandwidths exceeding 10 GHz. However, they tend to be expensive and suffer from low sensitivity (around -50 dBm) [5]. To address these limitations, narrowband systems, such as the superheterodyne receiver (SHR) are often considered. SHRs offer a narrow instantaneous bandwidth (around 100 MHz), which is shifted across the desired spectrum through a process known as superheterodyne mixing. This approach allows coverage of a wide frequency range (around 10 GHz) while maintaining high frequency selectivity and improved sensitivity (between -65 dBm and -85 dBm) [5],[6]. In addition, SHRs are simpler and more cost-effective than wideband alternatives. However, their main drawback is that only a small portion of the spectrum is observed at once. Consequently, signals transmitted in other frequency bands may go undetected. Improving the probability of intercepting emitters is therefore a central challenge for SHRs, especially when listening to radar emitters.

Indeed, radar systems often transmit short pulses over narrow bands and employ highly directional antennas. These systems scan sectors of angular space through either mechanical rotation or electronic beam steering; thus, the receiver is illuminated by the emitter only during short time windows. As such, intercepting radar signals with SHRs remains a significant technical challenge.

Two distinct approaches can be used to improve probability of intercept (POI). The first involves modifying the architecture of the hardware system to enable simultaneous listening to a wider portion of the spectrum. This can be achieved by deploying multiple synchronized SHR channels operating in parallel, each covering a different, nonoverlapping frequency band. The second approach focuses on optimizing the sequence and timing of the tuning of the listening band's center frequency, referred to as SHR scanning strategy optimization.

The first strategies used are typically based on periodic frequency sweeping [2]. In such schemes, the SHR

frequency is tuned either discretely or continuously through all frequencies in sequential order, from the lowest to the highest. Each frequency is periodically revisited, with a period (called revisit time) determined by the total width of the scanned spectrum. However, early research shows that these strategies may perform poorly in dealing with emitters exhibiting periodic emission [7]. In particular, when the receiver's revisit time is harmonically related to the emitter's scan period, a phenomenon known as "synchronization" occurs, characterized by systematic misses of the emitted signals. To address this issue, several mathematical models are developed to analyze and optimize the performance of deterministic strategies against one or multiple emitters [1], [2], [3]. The main statistics that are considered are the time before a first interception [intercept time (IT)] and the probability of intercepting during a given time (POI).

More recent scanning strategies use nondeterministic tuning sequences. This means that the order in which the frequency bands are chosen is determined according to a set of probabilities. In some cases, the duration of the observation on each frequency band (called dwell time) is also nondeterministic [8]. Some of these strategies are also adaptive, which means that they adapt the probabilities during an observation phase, according to previous interceptions [9], [10], [11], [12].

In this context, there is a need for a model that can predict the performance of random strategies in terms of IT and proportion of intercepted signals. This would allow not only to provide a theoretical baseline for evaluating adaptive strategies but also to optimize scanning parameters when some a priori knowledge is available. Another challenge is to account for complex emitters, such as low POI radars, which may present adaptive frequency hopping patterns. It is also important to consider the spatial scanning of emitters' antennas, which can be mechanical or electronic.

In our work, we develop a model that predicts IT and pulse interception ratio (PIR) of a randomly scanning multichannel SHR against spatially scanning and frequency-agile radar emitters. We apply this model to optimize a random scan against multiple known emitters. The rest of this article is organized as follows. In Section II, we briefly review state-of-the-art interception models. In Section III, we introduce the developed model along with mathematical interpretations. In Section IV, we examine the influence of key parameters on the predicted performance statistics, and in Section V, we compare our model with existing ones. In Section VI, we use a simulation environment to validate the model and explore its limitations. In Section VII, we apply the proposed model to optimize a random scan against multiple known emitters, with uncertainty intervals. Finally, Section VIII concludes this article.

II. RELATED WORK

In this section, we present an overview of the main methods for representing the interception problem for ES, as well as a summary of the existing models that aim to predict the interception performance for known configurations.

A. Representation of the Interception Problem

The first definition of the interception problem was proposed by Hatcher in 1976 [1]. He distinguished between detection (when a signal can be separated from noise) and interception (when the configuration between the emitter, receiver, and environment allows the reception of the emitted signal). An intercepted signal may remain undetected, whereas a detected signal must first have been intercepted.

Three interception scenarios are defined as follows.

- 1) *Beam-on-Beam*: Both the emitter and the receiver have moving directional antennas. In this case, an intercept occurs when the main lobes of both antennas are aligned.
- 2) *Beam-on-Frequency*: Either the receiver or the emitter has a moving directional antenna, while the other uses an omnidirectional antenna. The receiver scans the frequency spectrum, for example using superheterodyne reception. In this case, an intercept occurs when the main lobe of the directional antenna aligns with the other antenna while the receiver listens to the emission frequency.
- 3) *Frequency-on-Frequency*: The emitter uses an omnidirectional antenna, and either emits pulses or has a frequency-sweeping mechanism. The receiver also uses an omnidirectional antenna and scans the frequency spectrum. In this case, an intercept occurs when the receiver listens to a frequency band in which the emitter is active at this moment.

In all scenarios, Hatcher modeled the activities of the emitter and the receiver with periodic pulse trains, also called window functions. Each window function is characterized by its period T and window width τ .

Depending on the given scenario, T and τ correspond to different physical parameters, which are summarized in Table I.

Even though the initial work from Hatcher describes only three scenarios, the principle can be generalized by combining together multiple window functions to represent complex situations. For instance, a common scenario involves a pulsed radar emitter with a moving directional antenna, whose emissions are received by a digital SHR operating with a periodic strategy. This situation can be modeled using three window functions: one for the emission beam, one for the emitter's pulse train, and one for the periodic revisit of the emission frequency by the SHR. An intercept occurs only when all three window functions coincide. This particular situation is illustrated in Fig. 1.

Based on this representation, Hatcher also defined two interception statistics: 1) POI; and 2) IT.

- 1) POI is the probability of obtaining an interception in a given time window.
- 2) IT is the time needed to achieve a given POI.

Using this representation with the coincidence of window functions, various models are developed to predict the

TABLE I
Correspondence Between the Window Functions' Parameters and the Activity Types of the Emitter and the Receiver

Activity type	Window function period T	Window width τ
Receiver or emitter antenna rotation	Rotation period	Beamwidth / Rotation period
Frequency sweeping of SHR	Total bandwidth / Sweep rate	Observation bandwidth / Sweep rate
Frequency scanning of digital SHR	Revisit time	Dwell time
Pulsed emitter (e.g., radar emitter)	Pulse repetition interval (PRI)	Pulse width
Frequency sweeping of emitter	Total emission band / Sweep rate	1 / Sweep rate

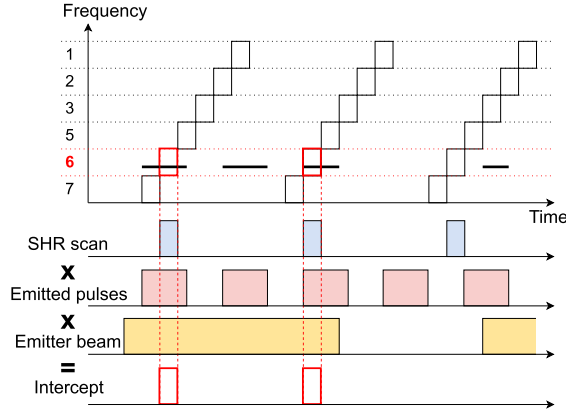


Fig. 1. Representation of the superheterodyne interception problem with pulse trains.

POI and IT in given situations. We will briefly describe them in the following subsection.

B. State-of-the-Art Interception Models

1) *Overview*: In the initial work from Hatcher [1], the theoretical values of POI and IT were computed for the case of two periodic window functions, by making the assumption of random and independent time shifts.

In their article, Self and Smith [2] extended the Hatcher's model and also questioned the assumption of independence of window functions following a first interception. Indeed, subsequent intercepts become deterministic and cannot be described by a model based on random time shifts. As a result, Self and Smith focused on modeling only the waiting time before the first intercept. They highlighted the issue of synchronization, which arises when the periods of the window functions are harmonically related. In such cases, IT tends toward infinity. However, their model does not account for this situation.

The first interception model accounting for synchronization was presented by Kelly et al. in 1996 [7], in the case of two window functions. They removed the random time shifts hypothesis and showed the presence of "holes" in the POI for specific period values. Through simulations, they showed that introducing jitter into the period of one of the window functions can reduce synchronization effects and bring the results closer to the predictions of the random time shifts model.

In the same year, Clarkson et al. [3] developed a new formulation of the interception problem of two window functions using Diophantine approximation, which is the

problem of approaching real numbers using series rational of rational numbers. By applying algorithms from this field, they provided a method to compute the waiting time before the first interception and the times before intercepting subsequent pulses. This algorithm was further developed in [13].

Two studies from 2006 and 2007 investigated the use of a random scanning strategy to alleviate the synchronization problem [8], [14]. The choice of the observation bands was made according to a continuous time Markov chain (CTMC). The authors derived an expression for maximum expected IT (MEIT) in the case of a periodic pulse train with zero pulse width. Their results showed that deterministic strategies can guarantee IT only when emitters are known with accurate intelligence. Otherwise, random strategies, such as the CTMC scan, proved more reliable.

In 2012, a study introduced a method for computing the POI for two or more window functions, based on cross-correlation [15]. The considered window functions can be deterministic or pseudo-random. This approach provides a POI statistic that is independent of the time shift between window functions, and without assumptions on their periodicity. However, it does not provide analytical expressions of IT.

As noted in [11] and [16], research literature on the subject of ES sensor scheduling is relatively limited, especially in recent years. The authors in the most recent works [9] and [11] used a different model of the interception problem involving transformed predictive state representations allowing for an online scanning strategy optimization, however, they do not provide explicit computations of interception statistics.

In the next parts, we describe in more details the following models: Self & Smith's model [2], Clarkson's model [13], and the CTMC model [4]. These will be compared to the model we have developed, in Section V.

2) *Self & Smith's Model*: The Self & Smith's model considers an arbitrary number of periodic window functions of periods T_i , with pulse widths of τ_i . Each window function represents a different condition on the occurrence of an interception. For instance, a first window function can represent the periodic illuminations of the emitter, a second can be the emitter pulse train, and a third can be the receiver scan which periodically revisits the frequency band where the emitter is active. In this case, the emitter is active in only one band. This method considers the revisit time of the receiver scan, so it is possible to consider multiple acquisition channels by simply dividing the revisit time by

the number of observing channels. Because of the assumption of random time shifts between all window functions, there are no constraints on the deterministic nature of the window functions. Thus T_i and τ_i can represent the mean periods and mean widths of random window functions. The Self & Smith's model determines the mean period between two intercepts as

$$\bar{t} = \frac{\prod_{i=1}^M (\frac{T_i}{\tau_i})}{\sum_{i=1}^M (\frac{1}{\tau_i})}. \quad (1)$$

To account for multiple overlaps between window functions, the maximum is taken over all combinations between the three window functions for each value of M [6].

3) *Clarkson's Model*: The Clarkson's model only considers two strictly periodic window functions, defined by their duty cycles. The first window function represents either the periodic illuminations or the periodic pulse train of a single frequency emitter. The second one represents the revisits of a sweeping SHR over the emitter's frequency band. Multiple acquisition channels can be taken into account by adjusting the duty cycle and period of the second window function accordingly. The Clarkson's model uses a generalized Farey series to compute recursively the ratios between the periods of the two functions which incur an infinite IT. It computes the maximum IT among all time shifts. We refer to the original article for implementation details on the algorithm for computing IT [13].

4) *CTMC Model*: The El Mahassni & Clarkson's CTMC model also incorporates two window functions. The first is strictly periodic with period T representing the emitter's periodic illuminations or the periodic pulse train. The second corresponds to a random receiver scan generated by a CTMC. In the frequency band of emission, the dwells are of random duration and are randomly spaced. At each instant, the probability of switching to another band follows an exponential distribution that decreases with time and resets upon each frequency switch. The decay rates associated with each individual band are stored in the rate matrix $\mathbf{Q}$, of size $(n_f \times n_f)$. Given a rate matrix, El Mahassni & Clarkson compute the mean IT in the case where the emitter's windows have a width of 0. The mean IT for each band corresponds to the diagonal elements of the following matrix:

$$\mathbf{F}(T) = \mathbf{X} \text{diag} \left\{ 0, \frac{T}{\pi_2(1 - e^{\lambda_2 T})}, \dots, \frac{T}{\pi_{n_f}(1 - e^{\lambda_{n_f} T})} \right\} \mathbf{X}^{-1} \quad (2)$$

where $\mathbf{Q} = \mathbf{X} \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_{n_f}\} \mathbf{X}^{-1}$ is the diagonalization of the rate matrix $\mathbf{Q}$, and π is the stationary distribution vector of the Markov chain.

Currently, to our knowledge, no existing model offers an explicit formulation of POI and IT in the case of a randomly scanning multichannel SHR, against spatially scanning and frequency-agile radar emitters. In the following section, we detail the model we developed to account for such situations.

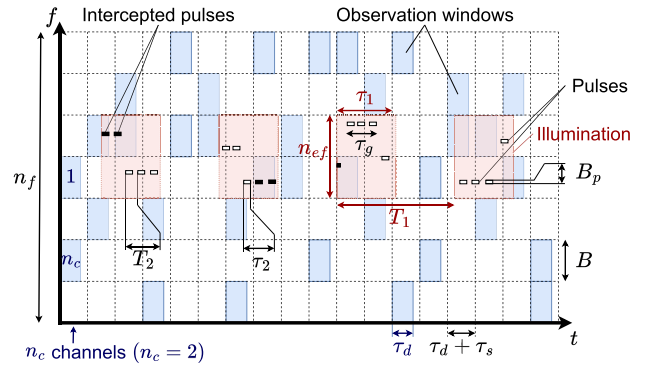


Fig. 2. Definition of the model parameters in the time-frequency domain.

III. INTERCEPTION MODEL

Our objective is to define and derive analytical expressions of three statistics associated with the superheterodyne interception problem in the case of a random scanning pattern: the POI in terms of pulses, the IT, and the PIR, which is the proportion of intercepted emitter pulses.

In Section III-A we define the main hypotheses and the validity domain of the model we propose. In Section III-B, we provide theoretical results that are useful for deriving the interception statistics. Then, in Section III-C, we describe the computation of the POI for one dwell, and extend the results to a given duration in Section III-D. In Section III-E, we derive the expression for IT, and finally for PIR in Section III-F.

A. Hypotheses and Validity Domain

We split the spectrum of interest into n_f discrete frequency bands of width B (Fig. 2). Observation windows, also called dwells, are assigned to one band at a time. Their duration τ_d (called dwell time) is fixed for a given observation sequence. We assume that there is a constant time gap τ_s between each dwell, called switch time. τ_s represents the time needed by the local oscillator of the superheterodyne receiver to be tuned to another frequency.

The superheterodyne scan is represented by n_c time-frequency observation windows at each time step (every $\tau_d + \tau_s$). Here, n_c denotes the number of superheterodyne acquisition channels available in parallel. At each time step, every channel is assigned to one of the n_f frequency bands. The bands are selected randomly using a uniform distribution, without replacement. Consequently, exactly n_c distinct frequency bands are observed at each time step.

We consider a pulsed radar emitter with a circularly rotating antenna. The receiver thus gets periodic illuminations from the emitter, with period T_1 . Each illumination has a duration of τ_1 , and is assumed to contain only the energy from the main lobe of emission. Inside each illumination, pulses are periodically emitted. Pulse width is noted as τ_2 and the period of pulses is T_2 . Pulses from the same illumination can take random frequency values inside a fixed range, defined by n_{ef} , the number of frequency bands in which pulses can be emitted. As frequency agility is not

TABLE II
Main Parameters of the Interception Model

Parameter name	Meaning
n_c	Number of acquisition channels
n_f	Total number of frequency bands
n_{ef}	Number of emitter frequency bands
T_1	Illumination repetition interval (s)
τ_1	Illumination time width (s)
T_2	Pulse repetition interval (s)
τ_2	Pulse time width (s)
τ_d	Receiver dwell time (s)
τ_s	Receiver switch time (s)
$\mathbf{p}_{ef}$	Proportion of emitted pulses per band
τ_g	Mean pulse group duration (s)
B_p	Pulse bandwidth (Hz)

necessarily uniform over all n_{ef} bands, we note $\mathbf{p}_{ef}$ the vector containing the proportion of pulses emitted in each band. Pulses can be also gathered into groups of same frequency. The mean duration of each group is noted τ_g . Each pulse has a frequency width B_p .

An intercept is considered to happen when at least an infinitesimal part of a pulse is located inside an observed frequency band during a dwell.

We assume that the initial time shift between the illuminations and the observation windows is random and uniformly distributed in $[0, T_1]$. Similarly, the initial time shift between the periodic pulses and the observation windows is random and uniformly distributed in $[0, T_2]$. It is important to note that the time shifts are fixed for a given observation sequence.

All the mentioned parameters are summarized in Table II

B. Theoretical Considerations

We first introduce a few preliminary theoretical results in order to compute the interception statistics in the context of the presented model. We will consider the coincidence between a single window and up to two periodic window functions, instead of the coincidence between two or three window functions only, as in existing models [2], [13]. Such results can be found in previous works [7], however, we provide here in-depth proofs and explanations that will be useful for deriving interception statistics. In Proposition 1, we determine the probability of coincidence of a periodic window function and a single window. We also extend the reasoning to determine the mean number of coincidences. In Proposition 2, we determine the probability of coincidence of two periodic window functions and a single window. In each case, we explain how the formulas are derived using geometrical representations, as this will be important for understanding further developments.

PROPOSITION 1 Let $\mathbf{A}(T_i, \tau_i)$ be a periodic window function with period T_i and a window width of τ_i . Let $\mathbf{B}(\tau_d)$ be a single rectangular function with width τ_d . Suppose there is a time shift θ between $\mathbf{A}$ and $\mathbf{B}$, which follows a uniform random distribution over the interval $[0, T_i]$. The probability

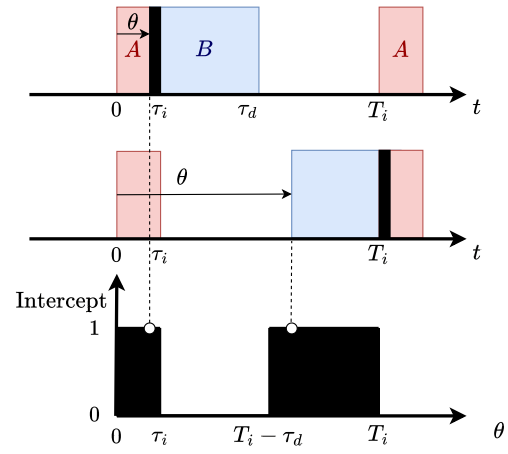


Fig. 3. Illustration of the computation of the probability of coincidence over one period.

$\mathbb{P}(\mathbf{A}, \mathbf{B})$ that $\mathbf{B}$ coincides at least partially with $\mathbf{A}$ is

$$\mathbb{P}(\mathbf{A}, \mathbf{B}) = \min \left(1, \frac{\tau_i + \tau_d}{T_i} \right) = \begin{cases} 1 & \text{if } \tau_d > T_i - \tau_i \\ \frac{\tau_i + \tau_d}{T_i} & \text{if } \tau_d \leq T_i - \tau_i. \end{cases} \quad (3)$$

PROOF Because the time shift θ is uniformly random, $\mathbb{P}(\mathbf{A}, \mathbf{B})$ corresponds to the proportion of time shifts that result in a coincidence. For each value of θ between 0 and T_i , we determine whether there is an overlap between $\mathbf{B}$ and at least one of two consecutive windows of $\mathbf{A}$. $\theta = 0$ is when $\mathbf{B}$ starts at the same time as one of the windows of $\mathbf{A}$, while $\theta = T_i$ is when $\mathbf{B}$ starts at the same time as the next consecutive window of $\mathbf{A}$ (see Fig. 3).

When $\tau_d > T_i - \tau_i$, there is always at least one overlap, whatever the value of θ , so coincidence happens on the range T_i . When $\tau_d \leq T_i - \tau_i$, two intervals imply an overlap. For $\theta \in [0, \tau_i]$, an overlap occurs on the first window of $\mathbf{A}$. An overlap on the second window of $\mathbf{A}$ occurs for $\theta \in [T_i - \tau_d, T_i]$. Thus, the sum of the ranges where coincidence happens is $\tau_i + \tau_d$, when $\tau_d \leq T_i - \tau_i$.

The proportion of time shifts implying a coincidence is determined by dividing the total range where coincidence happens by the range of all possible values of θ , which is equal to T_i . When $\tau_d \leq T_i - \tau_i$, it is equal to $\tau_i + \tau_d/T_i$ and it is equal to 1 otherwise. ■

It is important to note that the coincidence between $\mathbf{A}(T_i, \tau_i)$ and $\mathbf{B}(\tau_d)$ is equivalent to the coincidence between $\mathbf{A}(T_i, \tau_i + \tau_d)$ and $\mathbf{B}(0)$. In this case, $\mathbf{B}$ represents a window with an infinitesimal width.

Using this remark, we can derive the mean number of windows of $\mathbf{A}$ overlapping with $\mathbf{B}$, noted as $\overline{n_{\mathbf{A}, \mathbf{B}}}$

$$\overline{n_{\mathbf{A}, \mathbf{B}}} = \frac{\tau_i + \tau_d}{T_i}. \quad (4)$$

This corresponds to the number of equally spaced points (with a spacing of T_i) that fit within the interval $[0, \tau_i + \tau_d]$.

To introduce Proposition 2, we describe here a simplified geometrical representation of the coincidence between $\mathbf{A}(T_i, \tau_i)$ and $\mathbf{B}(\tau_d)$ (Fig. 4).

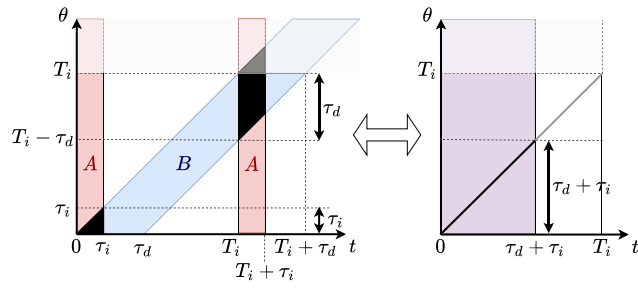


Fig. 4. Geometrical interpretation of the probability of coincidence (left) and its equivalent representation (right).

We use a representation of the different configurations parameterized by the time shift θ , by plotting the positions of **A** and **B** on a 2-D plane. The vertical axis represents the different time shifts θ , and the horizontal axis represents the time domain where the window functions are defined. **A** is represented as two parallel vertical bands of width τ_i . **B** is represented as a diagonal band, with a slope of 1 and a horizontal width of τ_d . Time shifts corresponding to a coincidence can be read by projecting on the vertical axis the regions where the bands overlap.

We now use the equivalence between $[\mathbf{A}(T_i, \tau_i), \mathbf{B}(\tau_d)]$ and $[\mathbf{A}(T_i, \tau_i + \tau_d), \mathbf{B}(0)]$ to simplify our representation (Fig. 4). **A** is represented as one vertical band of width $\tau_i + \tau_d$. **B** is represented as a diagonal line, with a slope of 1. In this configuration, time shifts implying a coincidence correspond directly to the region on the vertical axis where the line is inside the band. This means that a coincidence occurs when $\theta < \tau_d + \tau_i$.

PROPOSITION 2 Let $\mathbf{A}_1(T_1, \tau_1)$, $\mathbf{A}_2(T_2, \tau_2)$ be two periodic window functions, with $T_2 \ll \tau_1$. Let $\mathbf{B}(\tau_d)$ be a single rectangle function with a width of τ_d . Suppose there is a time shift θ_1 between $\mathbf{A}_1$ and $\mathbf{B}$, and a time shift θ_2 between $\mathbf{A}_2$ and $\mathbf{B}$. θ_1 and θ_2 are drawn from random uniform distributions on $[0, T_1[$ and $[0, T_2[$. The probability $\mathbb{P}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{B})$ that $\mathbf{A}_1$, $\mathbf{A}_2$, and $\mathbf{B}$ all coincide at least partially is

$$\mathbb{P}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{B}) = \min\left(1, \frac{\tau_1 + \tau_d}{T_1}\right) \times \min\left(1, \frac{\tau_2 + \tau_d}{T_2}\right). \quad (5)$$

Proof: We consider now the coincidence between three window functions $[\mathbf{A}_1(T_1, \tau_1), \mathbf{A}_2(T_2, \tau_2), \mathbf{B}(\tau_d)]$, as illustrated in Fig. 5. In the general case, this corresponds to the intersection of three couples of coincidence: $[\mathbf{A}_1(T_1, \tau_1 + \tau_d), \mathbf{B}(0)] \cap [\mathbf{A}_2(T_2, \tau_2 + \tau_d), \mathbf{B}(0)] \cap [\mathbf{A}_1(T_1, \tau_1 + \tau_d), \mathbf{A}_2(T_2, \tau_2 + \tau_d)]$. However, because we make the assumption that $T_2 \ll \tau_1$, it implies that we can neglect the amount of time shifts where $\mathbf{A}_1$ and $\mathbf{A}_2$ coincide with $\mathbf{B}$ but do not coincide between each other. The last couple is thus not considered here. By following the same reasoning as before, we know that a coincidence happens between the three functions when the following conditions are met:

$$C(\theta_1, \theta_2) = \begin{cases} \theta_1 < \tau_d + \tau_1 \\ \theta_2 < \tau_d + \tau_2. \end{cases} \quad (6)$$

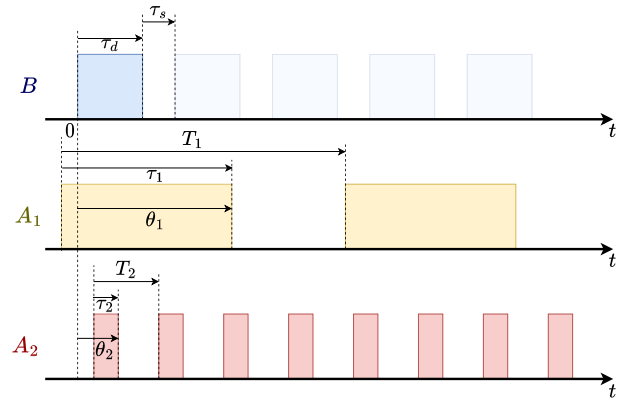


Fig. 5. Definition of the coincidence problem with two periodic window functions.

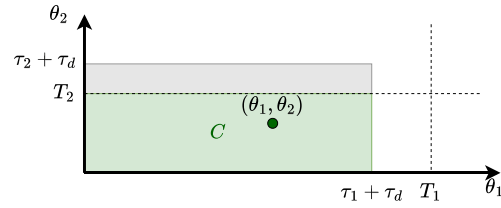


Fig. 6. Representation of the double coincidence condition, in the time shift plane.

These conditions can be represented graphically on the plane (θ_1, θ_2) as a rectangular region (Fig. 6). Because the time shifts have a uniform distribution, the probability that they fall into this region is the ratio between its area and the total area of the time shift domain. The probability of coincidence $\mathbb{P}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{B})$ can thus be expressed as

$$\mathbb{P}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{B}) = \frac{\min(T_1, \tau_1 + \tau_d) \times \min(T_2, \tau_2 + \tau_d)}{T_1 \times T_2}. \quad (7)$$

The min function is added to account for cases when the coincidence area exceeds the time shifts bounds T_1 or T_2 , i.e., the coincidence always happens for at least one of the periodic window functions. ■

C. POI for One Dwell

In this part, we determine the probability that a given dwell coincides with an emitted pulse. This means that the observation window must occur at the right time, and must be at the right frequency. We can thus divide this interception event into a temporal event and a frequency event.

We first want to know the probability of a temporal interception, which can be described by the following event:

I_t : “The observation window is located at a time when at least one illumination occurs and at least one pulse is emitted.”

Thanks to Proposition 2, if we model the illuminations, pulses and dwell by $[\mathbf{A}_1(T_1, \tau_1), \mathbf{A}_2(T_2, \tau_2), \mathbf{B}(\tau_d)]$, we know the probability $\mathbb{P}(I_t)$ of the event I_t

$$\mathbb{P}(I_t) = \frac{\min(T_1, \tau_1 + \tau_d) \times \min(T_2, \tau_2 + \tau_d)}{T_1 \times T_2}. \quad (8)$$

Then, we want to determine the probability of a coincidence in terms of frequency. This corresponds to the probability that at least one of the n_c acquisition channels is placed on at least one of the frequency bands where pulses are emitted, at a given time instant. We can define the event of frequency coincidence as the following.

I_f : “At a given time instant, at least one among the n_{ef} frequency bands of emission is observed.”

We can also define the following related event.

I_f^i : “At a given time instant, exactly i frequency bands of emission are observed.”

At a given time, the choice of the bands to observe follows a random uniform distribution, and the channels cannot be assigned to a same frequency band. This situation corresponds to an experiment where n_c among n_f bands are drawn without replacement, and among the n_f bands, only n_{ef} lead to a success. The probability of having i successes, which is $\mathbb{P}(I_f^i)$, follows a hypergeometric distribution, with parameters (n_c, n_{ef}, n_f) :

$$\mathbb{P}(I_f^i) = \frac{\binom{n_{ef}}{i} \binom{n_f - n_{ef}}{n_c - i}}{\binom{n_f}{n_c}} \quad (9)$$

with $\binom{N}{K} = \frac{N!}{K!(N-K)!}$. We can then derive the probability of the event I_f

$$\mathbb{P}(I_f) = 1 - \mathbb{P}(I_f^0) = 1 - \frac{\binom{n_f - n_{ef}}{n_c}}{\binom{n_f}{n_c}}. \quad (10)$$

We note that the event I_f does not represent exactly the reality of the frequency interception, in terms of pulses. Indeed, we made the implicit hypothesis that observing one of the n_{ef} bands of emission always implies a frequency interception of at least one pulse. However, it can happen that at a given time, pulses are emitted on other bands among the n_{ef} possible ones. Thus, the probability $\mathbb{P}(I_f)$ overestimates the probability of frequency interception in terms of pulses. To alleviate this, we can define the following event, corresponding to frequency interception in terms of pulses.

$\tilde{I}_f|I_t$: “Given that at least one pulse is temporally intercepted, at least one of them is inside an observation window.”

Its probability $\mathbb{P}(\tilde{I}_f|I_t)$ is equal to

$$\mathbb{P}(\tilde{I}_f|I_t) = 1 - \frac{\binom{n_f - \tilde{n}_{ef}}{n_c}}{\binom{n_f}{n_c}} \quad (11)$$

with $\tilde{n}_{ef}$ equal to

$$\tilde{n}_{ef} = n_{ef} - \sum_{b=1}^{n_{ef}} (1 - p_{ef_b})^{(1+\tau_d/\tau_g)(1+B_p/B)} \quad (12)$$

where p_{ef_b} is the proportion of emitted pulses in band b , B and B_p are the receiver and pulse bandwidths, respectively, and τ_g is the pulse group duration. The generalized binomial coefficient is used, like so: $\binom{z}{w} = \frac{\Gamma(z+1)}{\Gamma(w+1)\Gamma(z-w+1)}$, with z and w positive real numbers, and Γ the Gamma function. The derivation of these expressions is detailed in Appendix A1.

Finally, for a given time instant, the event of interception I can be defined as the intersection between the two events corresponding to frequency and temporal interception: $I = I_t \cap \tilde{I}_f$. Thus, the instantaneous POI is expressed as

$$\mathbb{P}(I) = \mathbb{P}(I_t) \times \mathbb{P}(\tilde{I}_f|I_t). \quad (13)$$

D. Global POI

We want now to determine the global POI, defined over a given time interval. First, this time interval is expressed as a number of observation time steps, denoted by k . The instantaneous POI corresponds to a trial defined for each observation time step. We repeat this trial k times and check the occurrence of the following event.

I_k : “The event I occurs at least once over the k observation time steps.”

We can first make the assumption that, for every time step, the time shifts θ_1 and θ_2 are drawn from random uniform distributions on $[0, T_1[$ and $[0, T_2[$. This corresponds to the random-phase theory, which has been previously developed [2]. In this case, the probability of the event I stays the same for all time steps. Therefore, this corresponds to a succession of independent Bernoulli trials with a parameter $\mathbb{P}(I)$. The probability to obtain at least one success is thus

$$\mathbb{P}(I_k) = 1 - (1 - \mathbb{P}(I))^k \quad (14)$$

which is the complementary of the probability to obtain no success within k steps. We can extend this expression to a continuous time interval t . We consider that an interception is only valid until the observation is completed. The extension to continuous time t is done by defining $k = \lfloor \frac{t}{\tau_d + \tau_s} \rfloor$, with $\lfloor \cdot \rfloor$ the floor function.

However, in reality, the assumption of independence between dwells is not exact. The time shifts θ_i are only drawn once and become deterministic for the following time steps. Suppose there is a time shift θ_i^0 between the observation window and one of the window functions at the first time step. Then, at time step k , the time shift becomes $\theta_i^k = \theta_i^0 - k(\tau_d + \tau_s) \bmod T_i$, where $\bmod$ denotes the generalized modulo function for real nonzero numbers: $x \bmod y = x - y \lfloor \frac{x}{y} \rfloor$. The modulo function accounts for the periodicity of the window function, allowing the time shift within one period only. If we replace θ_i with θ_i^k in (6), we obtain the conditions for a temporal coincidence occurring at the k th time step

$$C_k(\theta_1^0, \theta_2^0) = \begin{cases} (\theta_1^0 - k(\tau_d + \tau_s) \bmod T_1) < \tau_d + \tau_1 \\ (\theta_2^0 - k(\tau_d + \tau_s) \bmod T_2) < \tau_d + \tau_2. \end{cases} \quad (15)$$

In the time shift plane, these conditions mean that a point translated by $-k(\tau_d + \tau_s)$ from the initial point at (θ_1^0, θ_2^0) must lie within the rectangular region defined by (6), which has an area $(\tau_d + \tau_1)(\tau_d + \tau_2)$. When the translation implies that the point falls outside of the boundaries of the time shift domain, it is wrapped using the modulo function (see Fig. 7).

In order to find the probability of having at least one interception within k steps (i.e., I_k), we first need to find the probability of having no interception within k steps

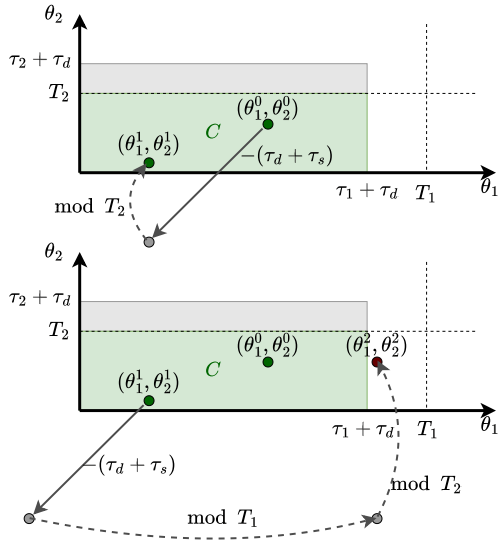


Fig. 7. Evolution of the double coincidence condition according to the considered time step. The left corresponds to the second step (a coincidence occurs), and the right corresponds to the third step (no coincidence occurs).

(i.e., $\bar{I}_k$). For given initial time shifts θ_1^0 and θ_2^0 , we first count the number $n_k(\theta_1^0, \theta_2^0)$ of temporal coincidences within k steps. Each time there is a temporal coincidence, there must be no frequency interception, which happens with a probability of $(1 - \mathbb{P}(\tilde{I}_f|I_t))$. Therefore, the probability of the event $\bar{I}_k$ given θ_1^0 and θ_2^0 is

$$\mathbb{P}(\bar{I}_k|\theta_1^0, \theta_2^0) = (1 - \mathbb{P}(\tilde{I}_f|I_t))^{n_k(\theta_1^0, \theta_2^0)} \quad (16)$$

with

$$n_k(\theta_1^0, \theta_2^0) = \sum_{j=0}^k \mathbb{1}_{C_j(\theta_1^0, \theta_2^0)} \quad (17)$$

and $\mathbb{1}_{C_j(\theta_1^0, \theta_2^0)}$ the indicator function equal to 1 when the conditions $\bar{C}_j$ are met. We can then average this probability over all time shifts to obtain $\mathbb{P}(\bar{I}_k)$

$$\mathbb{P}(\bar{I}_k) = \frac{1}{T_1 T_2} \int_0^{T_1} \int_0^{T_2} \mathbb{P}(\bar{I}_k|\theta_1^0, \theta_2^0) d\theta_2^0 d\theta_1^0. \quad (18)$$

In practice, integrals are replaced by sums over N discrete time shift values, which are evenly spaced between 0 and T_i . Finally, the probability of having at least one interception over k time steps is

$$\mathbb{P}(I_k) = 1 - \mathbb{P}(\bar{I}_k). \quad (19)$$

Similarly to the case of independent dwells (ID), we can extend this expression to a continuous time interval t by defining $k = \lfloor \frac{t}{\tau_d + \tau_s} \rfloor$.

E. Intercept Time

The notion of IT is closely related to the notion of POI. In fact, it corresponds to the time required to achieve a given POI. The time is counted from the first observation window to the end of the first observation window during which an interception occurs. We first consider a context with

discrete observation time steps. We write the discrete IT $k_I(\pi)$, with π as the POI to achieve. It is defined by the following condition:

$$\mathbb{P}(I_{k_I(\pi)-1}) \leq \pi \leq \mathbb{P}(I_{k_I(\pi)}). \quad (20)$$

We first consider the case of ID. By using (14), and after simplification, we obtain

$$k_I(\pi) \geq \frac{\log(1 - \pi)}{\log(1 - \mathbb{P}(I))} \geq k_I(\pi) - 1. \quad (21)$$

This expression is equivalent to

$$k_I(\pi) = \left\lceil \frac{\log(1 - \pi)}{\log(1 - \mathbb{P}(I))} \right\rceil \quad (22)$$

with $\lceil \cdot \rceil$ the ceiling function. In the case of dependent dwells (DD), there is no analytical expression for IT. However, an approximation can be found numerically by finding the point of zero-crossing of $\mathbb{P}(I_k) - \pi$.

We can extend this to determine the continuous IT $t_I(\pi)$, by multiplying $k_I(\pi)$ by the duration of the observation time step

$$t_I(\pi) = (\tau_d + \tau_s) \times k_I(\pi). \quad (23)$$

F. Pulse Interception Ratio

We want now to determine how many pulses are intercepted compared to the total number of emitted pulses. The mean proportion of intercepted pulses, which is the PIR, can also be interpreted in terms of probability. Indeed we can look at one emitted pulse and determine the probability of the following event.

I^P : “The pulse is located inside at least one observation window.”

The probability $\mathbb{P}(I^P)$ corresponds to the probability for each pulse to be intercepted, but we can also think of it as the PIR. To determine $\mathbb{P}(I^P)$, we can first look at the temporal aspect of pulse interception, by considering the following event.

I_t^P : “The pulse is located at an instant where at least one observation occurs, whatever the frequency band.”

We use Proposition 1 with $\mathbf{A}(\tau_d + \tau_s, \tau_d)$ the periodic function corresponding to the observation schedule, with a period of $\tau_d + \tau_s$ and a width of τ_d , and $\mathbf{B}(\tau_2)$ the rectangular function corresponding to the duration of one pulse, with a width of τ_2 . We get

$$\mathbb{P}(I_t^P) = \min(1, \frac{\tau_d + \tau_2}{\tau_d + \tau_s}). \quad (24)$$

Then, we want to determine the frequency aspect of pulse interception, using the following event.

$I_f^P|I_t^P$: “Given that the pulse is located at an instant where at least one observation occurs, at least one observation occurs in the frequency band of the pulse.”

We first consider the case where only one observation window is temporally present. The pulse spans over $n_p = 1 + B_p/B$ frequency bands, and n_c bands are observed without replacement among n_f bands in total. Similarly to (9) and (10), the probability of frequency coincidence

is equal to $1 - \frac{\binom{n_f - n_p}{n_c}}{\binom{n_f}{n_c}}$. To determine $\mathbb{P}(I_f^p | I_t^p)$, we need to consider k_p , the mean number of observation windows which are temporally present during the emission of one pulse. Using Proposition 2 with $\mathbf{A}(\tau_d + \tau_s, \tau_d)$ and $\mathbf{B}(\tau_2)$, we get a value of $\tau_d + \tau_2 / \tau_d + \tau_s$. However, because we consider that at least one observation is temporally present, this value cannot be less than 1: $k_p = \max(1, \frac{\tau_d + \tau_2}{\tau_d + \tau_s})$. The probability that at least one success occurs in the k_p trials is expressed as

$$\mathbb{P}(I_f^p | I_t^p) = 1 - \left(\frac{\binom{n_f - n_p}{n_c}}{\binom{n_f}{n_c}} \right)^{k_p}. \quad (25)$$

The global event of pulse coincidence is $I^p = I_t^p \cap (I_f^p | I_t^p)$. Therefore, we get the final expression for the PIR

$$\mathbb{P}(I^p) = \mathbb{P}(I_t^p) \times \mathbb{P}(I_f^p | I_t^p). \quad (26)$$

IV. INFLUENCE OF THE MAIN PARAMETERS ON THE MODEL

We describe in this section the influence of the model's parameters on the IT, as well as on the PIR. We consider one emitter, whose antenna scan period is $T_1 = 0.5s$, and illumination time width is $\tau_1 = 2.8$ ms. It emits pulses with a PRI $T_2 = 0.2$ ms, of length $\tau_2 = 1\mu s$, and of bandwidth $B_p = 10$ MHz. The frequency of the pulses is spread uniformly across $n_{ef} = 6$ bands of 50 MHz. Pulses form groups of same frequency with duration $\tau_g = 1$ ms. We set the receiver to have $n_c = 12$ acquisition channels, which listen to a total band of 5 GHz, which gives $n_f = 100$ frequency bands. The receiver switch time is $\tau_s = 0.1$ ms. The objective is to vary one parameter at a time to isolate the influence of each parameter. In all cases, we plot the analyzed statistic according to the dwell time τ_d , which ranges from 10^{-7} – 10^1 s. We use a logarithmic scale on both axes. We plot the statistic for four increasing values of the parameter under analysis.

A. Median IT

First, we plot $t_I(0.5)$, the median IT (MIT), with the variation of the following parameters: n_c , n_{ef} , τ_s . We will especially observe the evolution of the point of minimal MIT, as it is crucial for optimization.

In Fig. 8, we first consider fixed parameters, varying only the dwell time. The graph displays two curves. One uses the assumption of ID model when computing the POI, and the other considers DD.

We begin by comparing the global trends of both models according to dwell time values. When $\tau_d \rightarrow 0$, MIT seems to converge asymptotically toward a constant value. Then, between τ_2 and T_2 , the MIT drops until arriving at a minimum located at T_2 . Between τ_1 and T_1 , the MIT increases asymptotically toward the value at $\tau_d = T_1$. When $\tau_d > T_1$, MIT increases linearly with τ_d . Even though the shape of the MIT curve is transformed when varying the model parameters, we always observe that changes occur roughly for dwell times equal to τ_1 , T_1 , τ_2 , and T_2 .

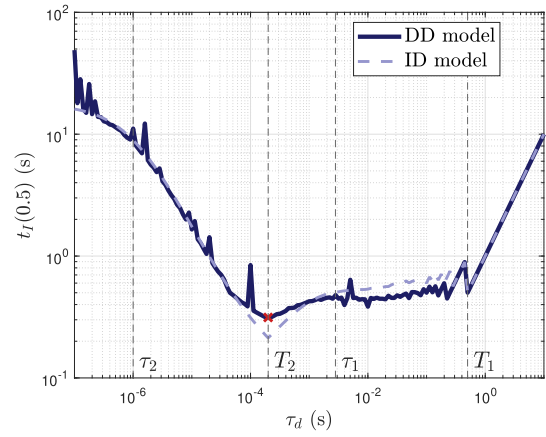


Fig. 8. MIT according to dwell time, as predicted by the proposed model, with (ID) and without (DD) the assumption of independence between dwells.

Even though both models have similar trends, important differences can be noted. The curve of the DD model is more flat in the region between T_2 and T_1 , and contains peaks at specific values of τ_d . In the region below T_2 , these peaks correspond to synchronization ratios between the observation windows and the pulses. For instance, a synchronization peak appears for $\tau_d = 0.1$ ms because in this case $\tau_d + \tau_s = T_2$, thus the period between two consecutive observation windows and the period T_2 are harmonically related, with a ratio of 1/1. For higher values of dwell time, the variations in MIT can be explained by the coarser discretization of time, which induces saw-tooth patterns when the values are expressed in continuous time.

We now choose $n_c = [1, 2, 10, 60]$ [Fig. 9(a)]. At first, MIT decreases similarly for all τ_d when n_c increases. When n_c becomes larger, the region between T_2 and T_1 progressively flattens, and the synchronization spikes are more visible. Indeed, when there are more observation channels, the frequency aspect of interception becomes less important than the temporal aspect.

Then, we choose $n_{ef} = [1, 2, 4, 10]$ [Fig. 9(b)]. We see that for $\tau_d \leq T_2$, the plots are mostly superimposed, which means that for dwell times smaller than the PRI, IT is independent of the width of frequency agility of the emitter. In this case, all emitters are equivalent to a single-band emitter. For $\tau_d > T_2$, we see a progressive decrease in IT when n_{ef} increases. As a consequence, the point of minimum MIT shifts toward larger values of τ_d . This corresponds to when multiple pulses can be observed on the same dwell. When dwell time becomes significantly larger than the PRI, many pulses are emitted during one dwell, thus all frequency channels of emission are likely to contain at least one pulse. As a result, a greater n_{ef} means a greater probability to achieve a frequency intercept and reduces IT.

We choose finally $\tau_s = [0, 1, 10, 100]\mu s$ [Fig. 9(c)]. When $\tau_s = 0$, the MIT decreases asymptotically toward around $T_1/2$ when $\tau_d \rightarrow 0$. This means that in the theoretical case where there is no switch time, an infinitesimally small dwell time implies an IT equal to the mean time of arrival of

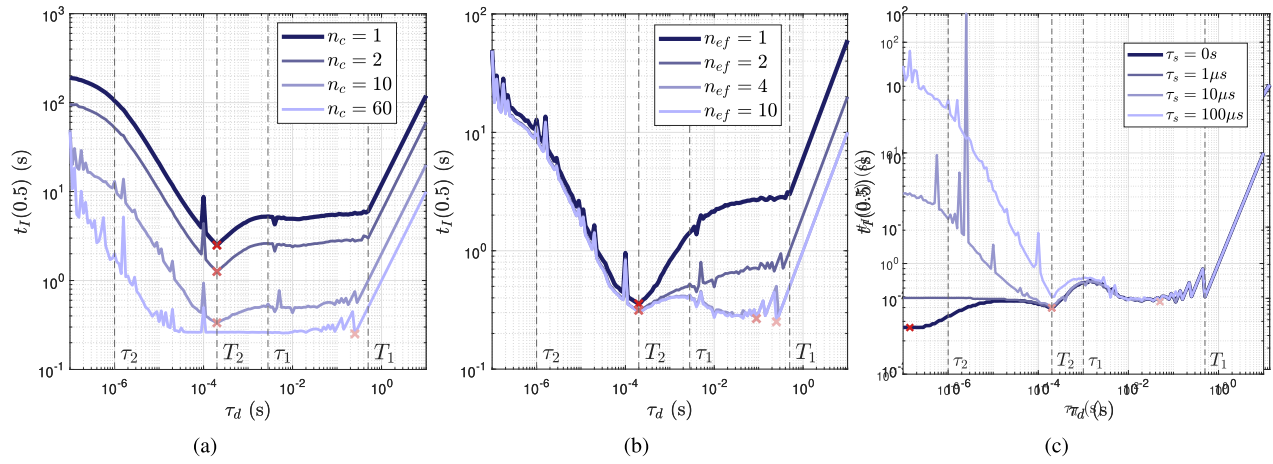


Fig. 9. MIT of the DD model according to dwell time, with increasing values of n_c (a), n_{ef} (b), and τ_s . (c) Red crosses indicate the points of minimum IT.

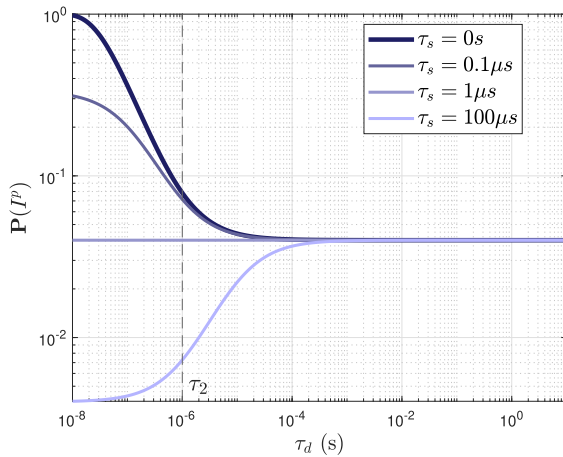


Fig. 10. Predicted PIR according to dwell time, with increasing values of τ_s .

the first illumination. When $\tau_s = \tau_2$, the MIT is flat below T_2 . In both the mentioned cases, no spikes are present, because a temporal intercept with the window function of the pulses always happens. For greater values of switch time, the limit in zero is proportional to τ_s . The spike patterns are heavily impacted by the value of switch time, which is expected, as τ_s plays a significant role in the synchronization phenomenon.

B. PIR

We plot the PIR according to τ_d , with variations of τ_s . We choose $\tau_s = [0, 0.1, 1, 100]\mu s$ (Fig. 10).

We first see that PIR is monotonous, with two asymptotic horizontal limits in 0 and $+\infty$. There are two points of local maximum curvature which are located roughly around $\tau_d = \tau_2$ and $\tau_d = \tau_s$. On these points, the PIR has opposite curvatures which are of the same magnitude. The curvature near τ_s is negative, whereas near τ_2 it is positive. Depending on which of these two parameters is greater than the other, the PIR will be either increasing or decreasing according to τ_d .

When $\tau_s = 0$, the PIR asymptotically tends toward 1 when $\tau_d \rightarrow 0$. This means that when there is no switch time, an infinitely small dwell time implies almost surely the interception of all pulses, whatever the emitter parameters. The limit in 0 decreases when τ_s increases. When $\tau_s = \tau_d$, the PIR is constant. When $\tau_s > \tau_2$, the PIR still has a horizontal asymptote, which indefinitely decreases with the increase in τ_s .

In summary, when the PRI is smaller than the receiver switch time, maximizing the PIR means choosing a dwell time as long as possible. On the contrary, when the PRI is greater than the switch time, maximizing the PIR means choosing an infinitesimally small dwell time.

V. COMPARISON WITH EXISTING MODELS

In this section, we compare our model with three state-of-the-art models, which explicitly derive IT statistics from emitter and receiver parameters in a superheteroyne interception context: 1) Self & Smith's model [2]; 2) Clarkson's model [13]; and 3) the CTMC model [8].

For all the following comparisons, we use a toy example of a single emitter with $T_1 = 3s$, $\tau_1 = 1s$, $T_2 = 0.2s$, $\tau_2 = 10ms$, $n_{ef} = 1$. The receiver has the following parameters: $n_c = 12$, $n_f = 100$, $\tau_s = 0s$, $\tau_d \in [10^{-4}, 10^1]s$. The total time of acquisition is supposed to be infinite.

A. Self & Smith's Model

We plot the MIT of our DD model as a function of τ_d , as well as the mean IT computed using the Self & Smith's formula with the same parameters (Fig. 11).

For dwell times above 20 ms, the curves have similar shapes, with a constant bias in logarithmic scale. This constant factor between the two models, which is approximately equal to 1.4, corresponds to the bias between the mean IT (for the Self & Smith's model) and the MIT (for the DD model). Indeed, because the IT values are distributed inside $[0, +\infty[$, the extreme values shift the mean upwards, contrarily to the median. Some variations due to the time

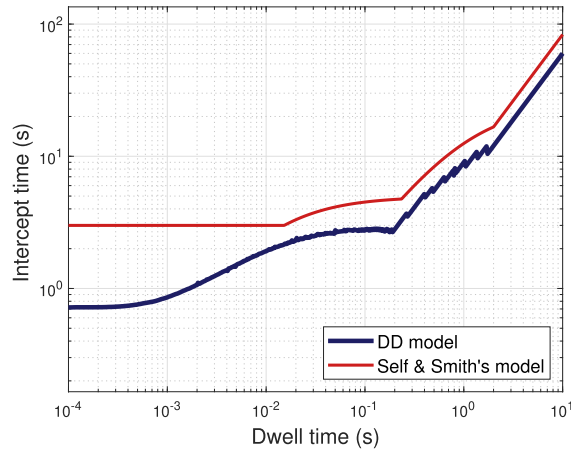


Fig. 11. IT according to dwell time, as predicted by the DD model (MIT) and the Self & Smith's model (mean IT).

discretization are also visible for the DD model for dwell times between T_2 and $T_1 - \tau_1$. Another visible difference is when τ_d approaches T_2 . The DD model has an inflection point at $\tau_d = T_2 - \tau_2$, whereas the other model has its inflection point slightly higher, for $\tau_d = \frac{\tau_1(T_2 - \tau_2)}{\tau_1 - (T_2 - \tau_2)}$. This could come from our assumption of deterministic periodic window functions to model the emitter.

When $\tau_d \rightarrow 0$ the DD model tends toward 0.716, whereas the Self & Smith's model stays at $\bar{\tau} = T_1$. This can be explained by the slight difference between the two definitions of IT. Indeed, Self & Smith define it as the mean period between two intercepts, which by definition cannot be lower than the illumination period. In our case, IT tends toward the median waiting time before a first emission of a pulse, given a random initial time shift.

B. Clarkson's Model

To compare with our model, we compute two versions of the Clarkson's model. The first considers only the pulses, by using the window function defined by T_2 and τ_2 . The second one considers only the illuminations, by using the window function defined by T_1 and τ_1 . Similarly, two versions of the DD model are computed: the "pulses" version is achieved by setting $\tau_1 = T_1$, and the "illumination" version is made by setting $\tau_2 = T_2$ (Fig. 12).

Clarkson's IT shows numerous peaks indicating when synchronization occurs. In both cases, the peaks occur in an area, where $\frac{\tau_i}{n_f/n_c - 1} < \tau_d < T_i - \tau_i$. Indeed, when the revisit time ($n_f \tau_d / n_c$) becomes smaller than the window width (τ_i), and when τ_d becomes greater than the gap between two windows ($T_i - \tau_i$), no synchronization can occur, as it is impossible to miss an emitted pulse. The MIT computed by our model in both cases constitutes a lower bound to their sequential counterpart, with a margin of around 1.4. This margin corresponds to the same bias as with the Self & Smith's model. The inflection points at $\tau_d = T_i - \tau_i$ are equal. Similarly to the Self & Smith's model, the IT cannot fall below T_i , since in the worst-case scenario, one must wait an entire period before an intercept becomes possible.

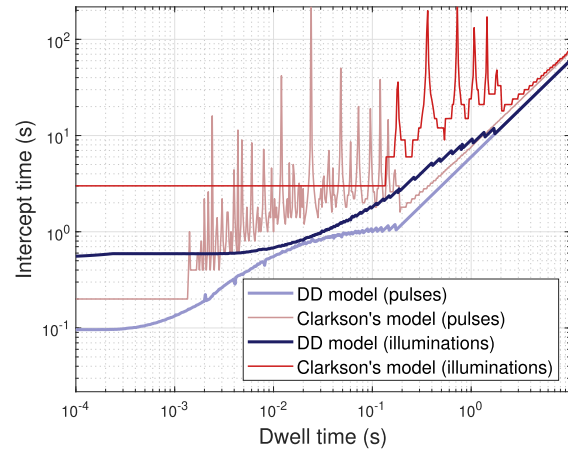


Fig. 12. IT according to dwell time, as predicted by the DD model (MIT) and the Clarkson's model (maximum IT). Either pulses or illuminations are considered.

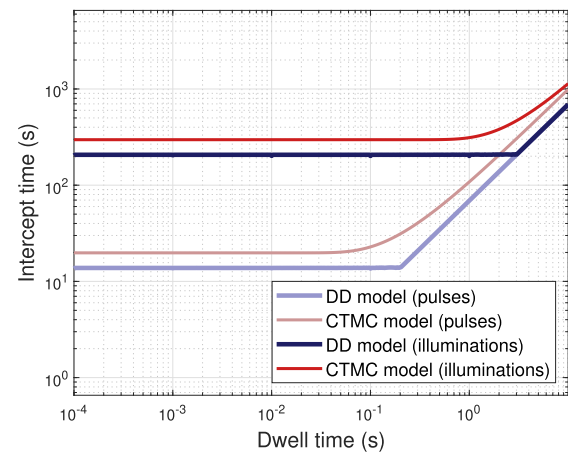


Fig. 13. IT according to dwell time, as predicted by the DD model (MIT) and the CTMC model (mean IT). Either pulses or illuminations are considered.

C. CTMC Model

In the same way as with the Clarkson's model, we use two configurations, one where we consider illuminations only, and one where we consider pulses only (Fig. 13). Because multiple acquisition channels are not accounted for in the CTMC model, we set $n_c = 1$.

For the sake of comparison, we choose a rate matrix $\mathbf{Q}(\tau_d)$ so that each band is equally probable and the mean dwell time on each band is equal to τ_d . Its diagonal elements are all equal to $-1/\tau_d$ and all other elements are equal to $1/\tau_d(n_f - 1)$. It corresponds to the optimal matrix $\mathbf{Q}^*$ found in the original research paper [8].

The IT of the CTMC model has in both cases two asymptotes: when τ_d tends toward infinity, IT tends toward $n_f \tau_d$, and when τ_d tends toward 0, IT tends toward $n_f T_i$. The asymptotes of the DD model are similar, with a bias of around 1.4; however, the evolution of IT is purely linear. The horizontal part is explained by the fact that, when $\tau_d < T_i$, a smaller dwell time implies more revisits on the emitter's band, but also a lower probability of landing on a pulse (or

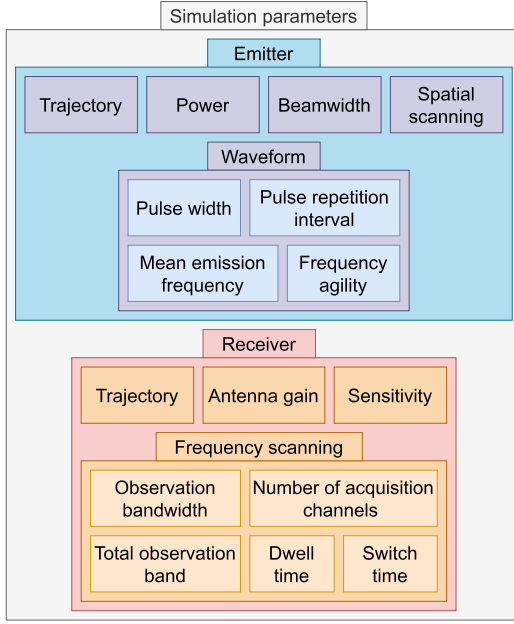


Fig. 14. Simulation parameters, grouped by categories.

illumination) at each try. These two effects compensate each other, which gives a constant IT. When $\tau_d > T_i$, an intercept always occurs when the emitter's band is observed. Thus, in this case, IT depends only on the revisit time on the emitter's band, which is $n_f \tau_d$. The same reasoning applies for the CTMC model, however, because the dwell time is random and only its mean is fixed, the transition between the two asymptotes is smooth.

VI. COMPARISON WITH SIMULATION

A. Presentation of the Simulation

To validate the proposed model, we compare the IT and PIR statistics with simulation results. The parameters used in the simulation are summarized in Fig. 14.

Based on these parameters, we simulate the trajectories of the emitter and receiver, along with the direction of their antennas. We use the radar equation to compute the received power for each emitted pulse. If the signal-to-noise ratio (SNR) is greater than a defined minimum SNR and the receiver is located within the main lobe of the emitter (defined by the half-power beamwidth), we consider the pulse as a candidate for being intercepted. We do not account for the effects of the environment on wave propagation; only the noise level is considered.

B. Parameters Used

We now describe the chosen parameters for the simulated example we use in this section. The receiver is static in space with an antenna gain of 2 dB, a minimum SNR required of 20 dB, and the following frequency scanning parameters.

- 1) Observation bandwidth $B = 50$ MHz.

TABLE III
Main Parameters of the Simulated Emitters

Emitter	1	2	3
Center frequency	3.4 GHz	2 GHz	1 GHz
Frequency range	0	150 MHz	100 MHz
θ_{3dB}	0.8°	1.5°	3.3°
n_{ef}	1	4	3
τ_1	5.6 ms	12.5 ms	45.8 ms
T_1	2.5 s	3 s	5 s
τ_2	0.25 μ s	2 μ s	10 μ s
T_2	0.6 ms	0.04 ms	1.2 ms

- 2) Total observation band between 25 and 5025 MHz, which gives $n_f = 100$.
- 3) Switch time $\tau_s = 10$ μ s.
- 4) Number of acquisition channels $n_c = 4$.
- 5) Dwell time $\tau_d \in [10$ μ s, 1 s].

The noise spectral density at the receiver is set to -100 dBm/Hz.

Main emitter parameters are summarized in Table III. In this case, the emitted pulses are uniformly distributed in frequency in a given range around their center frequency. When related to the receiver bandwidth, this gives the number of emitted frequency bands n_{ef} . An additional band arises from the frequency shift of the observation windows equal to 25 MHz in the simulated example. Each pulse has a different frequency value within the n_{ef} bands ($\tau_g = \tau_2$), and is not modulated, i.e., $B_p = 0$.

All emitters have a circular scanning antenna. The illumination width corresponds to $\tau_1 = \frac{\theta_{3dB} T_1}{360^\circ}$, with θ_{3dB} the half-power beamwidth in azimuth.

Emitters all have an emission power of 30 kW and a maximum gain of 30 dB. The emitters are static in space and are located within the maximum range of the receiver, which is approximately 10 km for this configuration.

The duration of each simulation episode is set to 30 s. The statistics are averaged over 200 episodes for each configuration. 125 different values of dwell time are used, which are evenly spaced on a logarithmic scale between 10 μ s and 1 s. For each configuration, we measure the median and 99th percentile of IT, as well as the PIR. A different random time shift is applied to the illuminations and the pulses at each episode. The simulated pulses are represented in Fig. 15, where each dot is a different pulse that arrives at the receiver.

C. Metrics

To measure the accuracy of our model compared to simulated results, we use two metrics. For the PIR, we use the median absolute error (MdAE), and for the IT, we use the median symmetric accuracy (MSA).

The MdAE has the following expression:

$$\text{MdAE}(y, \hat{y}) = \text{med}(|\hat{y}_i - y_i|) \quad (27)$$

where y_i corresponds to each measured value, $\hat{y}_i$ corresponds to each predicted value, and $\text{med}(\cdot)$ is the median function. We choose the MdAE metric for the PIR because of its interesting properties [17] described as follows.

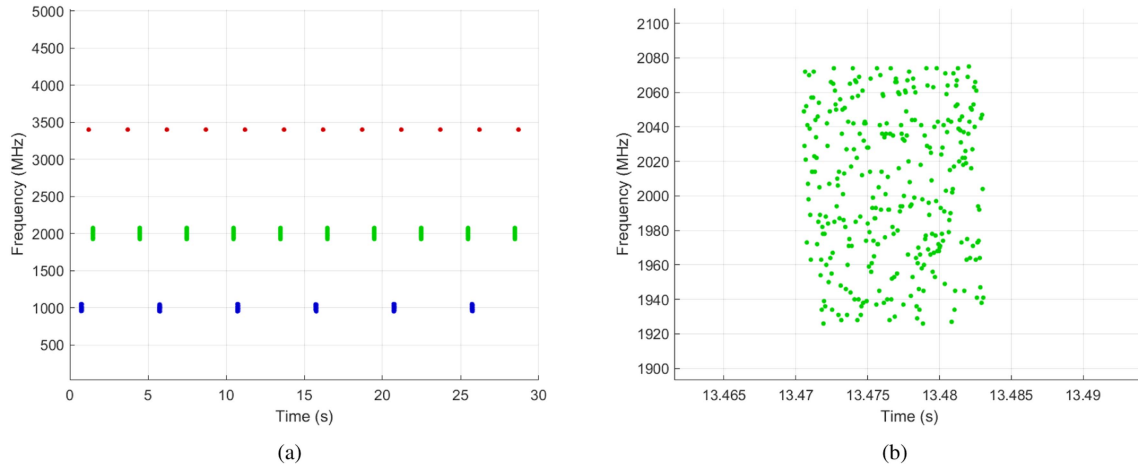


Fig. 15. (a) Time-frequency representation of the simulated pulses arriving at the receiver. (b) Zoom on one illumination, which shows the individual pulses.

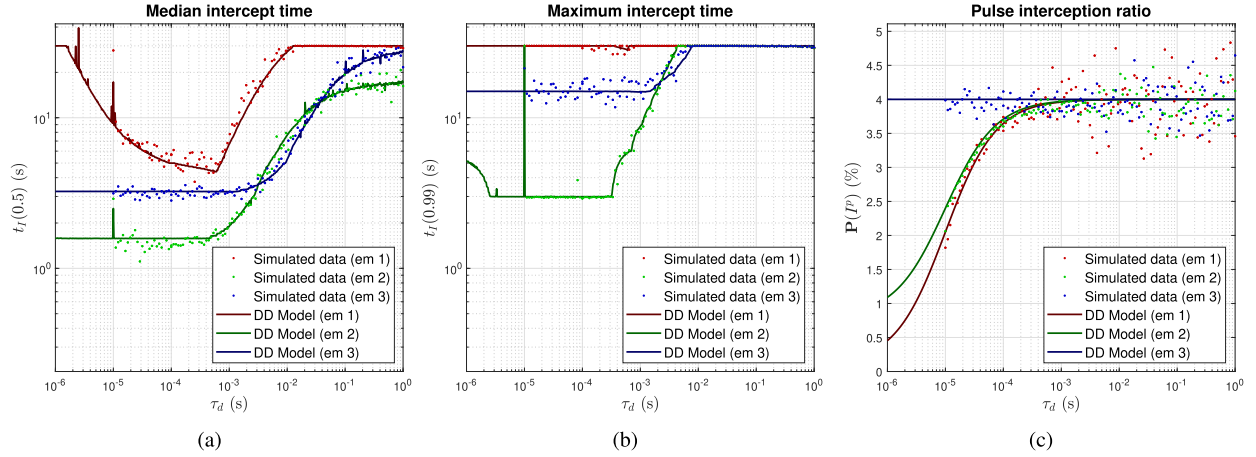


Fig. 16. (a) MIT, (b) maximum IT, and (c) PIR, from simulation data and predicted by the DD model.

- 1) It is expressed in the same units as the data.
- 2) It is robust to outliers and handles zero values.
- 3) It is symmetric, i.e., the positions of predictions and measures can be switched without changing the result.

The MSA has the following expression:

$$\text{MSA}(y, \hat{y}) = 100 \times (\exp(\text{med}(|\log(\hat{y}_i/y_i)|)) - 1). \quad (28)$$

For the IT, we prefer to use the MSA metric, because it is not only robust to outliers and symmetric but also a relative error. This means that it does not depend on the scale of the data, since the error is expressed as a ratio rather than a difference. As IT generally spans several orders of magnitude, a relative error is well adapted. Compared to other common relative error metrics, MSA penalizes overprediction and underprediction equally [17]. The MSA metric is expressed as a percentage and can be interpreted as the value so that 50% of the unsigned percentage error for each data point is defined as $\eta_i = 100 \times \frac{\max(\hat{y}_i, y_i) - \min(\hat{y}_i, y_i)}{\min(\hat{y}_i, y_i)}$.

TABLE IV
Precision Metrics of the DD Model Compared to Simulated Data, Within Its Validity Domain

Emitter	Median IT	Maximum IT	PIR
1	8.61%	6.69%	0.149
2	6.42%	1.49%	0.088
3	5.29%	11.61%	0.147

D. Validation Within the Validity Domain

In this part, we evaluate whether our model accurately predicts IT and PIR within its domain of validity, i.e., when the stated assumptions are satisfied. We first plot the simulation results and model predictions according to dwell time for median and maximum IT (short for 99th percentile IT) and PIR (Fig. 16). The range of dwell time is extended to $[10^{-6}, 1]$ s for the model prediction only, to facilitate further interpretation. For IT, we compute only the DD model. The number N of discretized time shifts used in the DD model is set to 50. The IT values exceeding the maximum simulation duration (30 s) are capped at this value, both in the simulation results and the model predictions.

TABLE V
Precision Metrics of Existing Models Compared to
Simulated Data

Emitter	Self & Smith	Clarkson	CTMC
1	7.76%	246.7%	5396%
2	108.81%	2469%	1567%
3	30.40%	576.3%	1341%

For IT metrics [Fig. 16(a) and (b)], we observe that the prediction curve of the DD model closely follows the trend of the simulation data across all time scales. The model accurately predicts the spike at $\tau_d = \tau_s$ for emitters 1 and 2, which results from temporal synchronization between dwells and emitted pulses. The DD model also captures the constant part of maximum IT for emitter 2, which is equal to the illumination period.

For the PIR [Fig. 16(c)], the model seems to closely follow simulation data in the constant region, as well as when the PIR drops for emitters 1 and 2.

We use the previously described metrics to quantitatively evaluate the model predictions. They are computed only over nonsaturated data, i.e., data values below 30 s, and are all summarized in Table IV.

For MIT, all MSA values stay below 10%, whereas for maximum IT it stays under 12% for all emitters. The MdAE of the PIR stays below 0.15 for all emitters. This confirms quantitatively that the DD model predicts IT and PIR with good accuracy.

In conclusion, we used examples of three complex emitter/receiver configurations, which fit within the assumptions of the model. For the considered variables, the trends according to τ_d of simulated and predicted data are similar. Metrics of relative error for IT measures are contained under acceptable bounds for all scales of dwell time within the considered range. Observed errors can be mainly explained by the variability of simulation data. PIR measures are also well fitted by the model predictions.

E. Comparison With Existing Models

In this part, we apply the three existing models presented in Section V on the simulated example, and measure their accuracy when computing IT, using the MSA metric. For Self & Smith's and the CTMC models, we compare their results on the basis of the statistical average of IT among all episodes. We use the "illuminations" setup for the CTMC model. The results of the Clarkson's model are compared to the maximum IT among all episodes. To account for the double periodicity, we use the maximum between the "pulses" and "illumination" configurations. All results are summarized in Table V.

For the Self & Smith's model, we see that emitter 1, which has no frequency agility, is the only one whose error stays under 10%. For the others, larger agility bands imply larger errors. Indeed, frequency agility is not considered in this model.

For the Clarkson's model, we see that no emitter is correctly predicted. Indeed, due to the many predicted synchronization spikes that would occur in a sequential scan,

TABLE VI
Precision Metrics of the DD Model Compared to
Simulated Data, With Random PRI Emitters

Emitter	Median IT	Maximum IT	PIR
1	9.41%	7.55%	0.133
2	6.83%	1.82%	0.076
3	5.71%	8.63%	0.106

the error increases significantly, above 100%. The error increases even more with the width of frequency agility.

For the CTMC model, the error is even larger, for all emitters (above 1000%). Because its assumptions (single pulse train, zero-width pulses, variable dwell time) are too far from the actual scenario, it fails to capture the evolution of IT.

In conclusion, we can say that for a scenario of frequency agile emitters intercepted using a random scan, the DD model is significantly superior to existing ones for predicting IT.

F. Extensions to Complex Periods

We now want to see if the assumption of strictly periodic window functions for the emitter can be relaxed. This can either mean that the PRI is not constant, or that illuminations come from more complex antennas, such as electronically scanned arrays (ESA) [18]. In such cases, T_1 and T_2 could represent the mean interval between two illuminations or pulses.

We will see first if the model predictions are still relevant in the case where PRI follows a random uniform distribution between τ_2 and $2T_2 - \tau_2$. All other parameters are kept the same. In this case, T_2 is the mean PRI. We keep the same model prediction; however, we run the simulation again with the random PRI emitter.

Simulation data and model prediction are plotted in Fig. 17. We see that simulation data with a random PRI emitter is very similar to the periodic case, with some slight differences. Indeed, the spikes predicted by the DD model are absent from simulation data. This is expected, as the synchronization phenomenon disappears when the window functions have nondeterministic periods [14].

MSA and MdAE scores for the random PRI emitter are summarized in Table VI. In summary, quantitative results show that the model predicts IT and PIR with the same accuracy as with the periodic PRI emitter, except for the synchronization spikes which are wrongly predicted.

We now consider the case where the illuminations are no longer strictly periodic. For example, a radar emitter equipped with an electronically scanned array can rapidly generate beams over a wide angular sector, using complex schedules. These schedules may depend on the environment of the radar emitter, such as the presence or absence of nearby threatening targets. From the receiver's point of view, we model these emitters by illuminations with a random uniform period between τ_1 and $2T_1 - \tau_1$. The illumination width τ_1 is kept constant. As in the previous section, we retain the same emitter parameters, changing only the constant illumination periods to random ones.

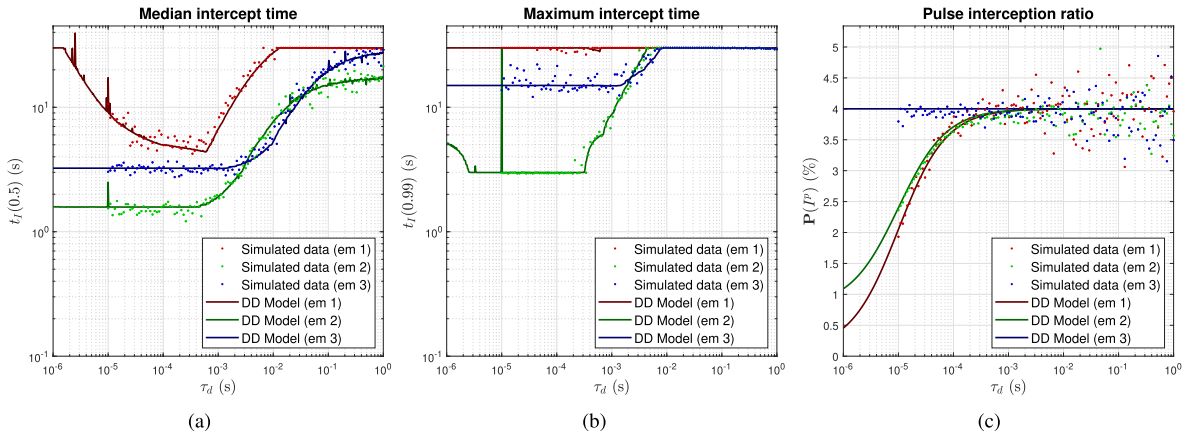


Fig. 17. (a) MIT, (b) maximum IT, and (c) PIR, from simulation data (with random PRI) and predicted by the DD model.

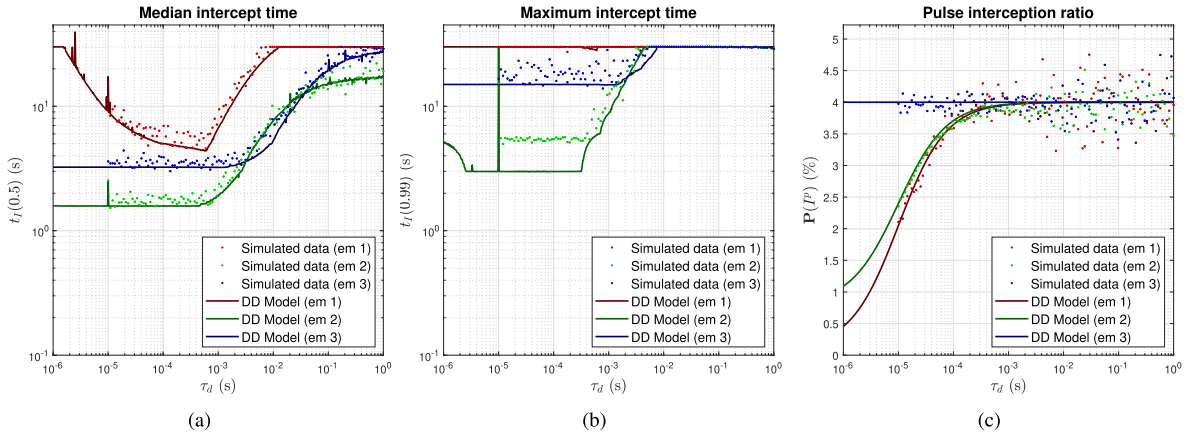


Fig. 18. (a) MIT, (b) maximum IT, and (c) PIR, from simulation data (with random illumination periods) and predicted by the DD model.

TABLE VII
Precision Metrics of the DD Model Compared to
Simulated Data, With Emitters With Random
Illumination Periods

Emitter	Median IT	Maximum IT	PIR
1	14.21%	//	0.116
2	8.78%	75.29%	0.096
3	10.39%	21.32%	0.105

Simulation data and model prediction are plotted in Fig. 18. For MIT [Fig. 18(a)], we see that simulation data are systematically superior to the DD model predictions. The global trend as well as the spiking patterns are still preserved. On the maximum IT graph [Fig. 18(b)], this bias is more significant. For emitter 1, all simulation data are saturated at 30 s, so no interpretation can be made. For emitter 2, we can see that the constant region is far higher than what the model predicts, at around 5.5 s instead of 3. This value corresponds roughly to the 99th highest illumination period among all the randomly chosen ones. The PIR values [Fig. 18(c)] remain unchanged; thus, the model still fits simulation data.

MSA scores are significantly higher than the standard case (see Table VII). This confirms the presence of the bias on IT introduced by the random illumination periods.

The error metrics for PIR are still similar to the standard case.

On one hand, we see that our model can also account for random PRI values without significant loss of precision. We can note, however, that the synchronization spikes on IT disappear compared to the standard case. On the other hand, random illumination periods have a more significant impact on the prediction performance of IT, although the global trends and spiking patterns are preserved. The PIR prediction is correctly predicted in all cases. Thus, we can conclude that our model can account for random PRI emitters and, to a given extent, to emitters with complex illumination schedules, such as ESA. Indeed, according to [18], the illumination patterns produced by ESA with adaptive scheduling can be approximated using either a periodic window function, or a fully random one, in terms of IT.

VII. MINIMAX OPTIMIZATION

We now consider the case where we have some information about some emitters which are likely to appear in a given operational context. We describe in this section some ways to use our model to optimize the interception of these emitters in particular.

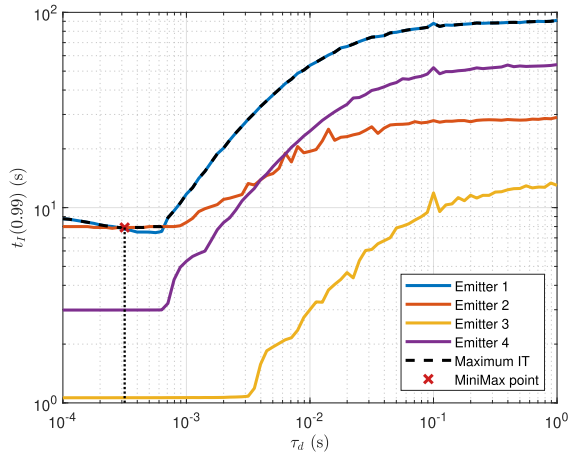


Fig. 19. Model predictions of maximum IT for the four listed emitters, according to dwell time.

A. Method Description

We use a method similar to Clarkson's minimax optimization approach [13]. This method consists in comparing the IT of all known emitters according to the receiver revisit time, and selecting the configuration that minimizes the maximum IT across all emitters. Clarkson applies this method using its model, that computes the maximum IT for a sequential superheterodyne scan. Only the illuminations are considered in his approach, with a correction applied to the beamwidth to account for the PRI. The main advantage of the minimax optimization is that it guarantees intercepts for all listed emitters within the minimum maximum IT found by the method. Moreover, the optimization is independent of the emitter's operating frequencies. However, it can be highly sensitive to errors in the revisit time setting. In Clarkson's example, a relative tolerance of only 0.0021% on the revisit time leads to an increase of at least 25% of IT. Errors on the parameters of the listed emitters can also imply large deviations from the effective optimal point, considering the presence of numerous peaks near areas of minimal IT.

We now want to use a similar optimization scheme with the model we have developed. Since we are using a random scan, there is no explicit maximum IT, as it is theoretically infinite. However, we can work with the IT corresponding to a high POI, 99% for instance. For each emitter in a given list, we assume that its parameters are precisely known, namely n_{ef} , τ_1 , T_1 , τ_2 , and T_2 . We can then compute $t_I(0.99)$ for all emitters according to the dwell time $\tau_d \in [100\mu s, 1s]$, and find the value of τ_d that minimizes the maximum IT among all listed emitters. Because IT for random scans has smaller variations according to dwell time compared to sequential scans, errors in setting dwell time do not greatly affect the overall performance.

However, we can anticipate that errors on the parameters of the listed emitters may lead to a deviation from the actual optimal point. Indeed, as observed in Section IV, variations in each parameter can cause either an increase or a decrease in IT, which may in turn shift the point of minimum IT.

TABLE VIII
Example List of Emitter Parameters

Emitter	n_{ef}	τ_1	T_1	τ_2	T_2
1	1	7 ms	2.5 s	0.25 μs	600 μs
2	5	5 ms	4 s	0.1 μs	250 μs
3	3	38 ms	1.1 s	0.7 μs	30 μs
4	2	20 ms	3 s	2 μs	40 μs

We assume that the variation of IT with each parameter is monotonous, which means that the higher and lower bounds of IT are located at the higher and lower bounds of the parameter. For instance, if $T_i \in [T_i^l, T_i^h]$ ($i = \{1, 2\}$), then $t_I(\pi, T_i^l) \leq t_I(\pi, T_i) \leq t_I(\pi, T_i^h)$. In this case, the order of the bounds is kept. It can also be reversed. Indeed, if $\tau_i \in [\tau_i^l, \tau_i^h]$ ($i = \{1, 2\}$), then $t_I(\pi, \tau_i^h) \leq t_I(\pi, \tau_i) \leq t_I(\pi, \tau_i^l)$. It is also the case for n_{ef} . With this in mind, we can use the minimax optimization method over intervals of parameters instead of single values. This allows us to obtain an upper bound on IT, even for incorrect values of some parameters of the emitter list.

To define these intervals, we use the same logarithmic relative error ϵ for all time-related parameters, namely τ_1 , T_1 , τ_2 , and T_2 . We define $\epsilon = \log_{10}(\hat{y}/y)$, where $\hat{y}$ is the estimated value of the parameter and y the true value. An error of 1 indicates that the estimated value is one order of magnitude greater than the true value, while an error of -1 means that it is one order of magnitude lower. For n_{ef} , we use a standard error defined as $\eta = \hat{y} - y$. In a more realistic context, it could also be possible to manually define intervals for each parameter of each emitter, depending on the confidence in the emitter information.

B. Example Use on Four Emitters

For testing the method, we use an example list of four emitters (see Table VIII). Their parameters are loosely based on typical military radar characteristics of naval, terrestrial, and aerial units. Pulses are uniformly spread in the n_{ef} bands, and are not modulated. The receiver parameters are: $n_f = 100$, $n_c = 12$, $\tau_s = 10 \mu s$, and $\tau_d \in [10^{-4}, 1] s$.

We first use the listed parameters as input to the minimax optimization method. The resulting curves are shown in Fig. 19. The minimax point is located at $\tau_d = 310 \mu s$, for an IT of around 7.8 s. This means that for this value of dwell time, there is a 99% chance that all emitters are intercepted within 7.8 s. In this case, only emitters 1 and 2 are used to determine the optimal point, as they are the most difficult to intercept in the list. To compare with Clarkson's optimization in terms of error tolerance in setting dwell time, we note that a 25% increase in IT is achieved by changing τ_d by a factor of around 3. This means that random scan optimization is far less sensitive to dwell time errors than sequential scan optimization.

Then, we set a logarithmic relative error $\epsilon = 20\%$ on all time-related parameters (τ_1 , T_1 , τ_2 , T_2), to compute higher bounds on IT. For n_{ef} , we only use the worst case, i.e., $n_{ef} = 1$. Real IT as well as worst-case IT are plotted in Fig. 20. The worst-case minimax point is located at $\tau_d = 430 \mu s$, for an IT of around 29 s. This means that

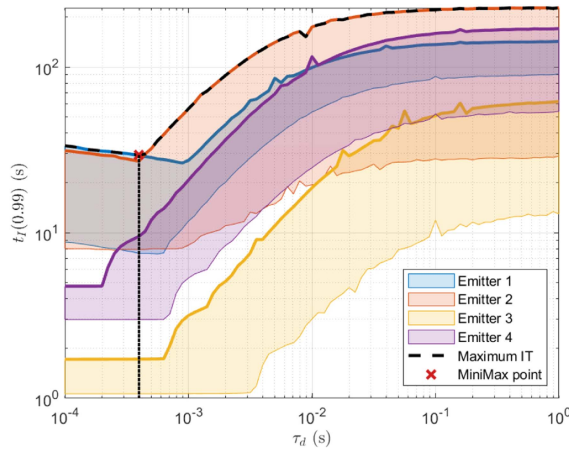


Fig. 20. Model predictions, with uncertainty intervals, of maximum IT for the four listed emitters, according to dwell time.

there is a 99% chance that all emitters are intercepted within 29 s, even if all parameters are incorrect by a factor of $10^{0.2} \approx 1.58$. Using the worst-case minimax point with the real parameter values induces an error on IT of around 77 ms, whereas using the standard minimax with worst-case parameter values induces an error on IT of around 2.2 s. This illustrates the idea that the worst-case approach allows us to reduce the risk of IT increase.

VIII. CONCLUSION

In the context of ES, we aim to optimize the frequency scanning parameters of a multichannel superheterodyne receiver by exploiting prior information about the emitters in a given operational situation. We focus on random frequency scanning strategies with constant dwell time on each band.

We first develop a probabilistic mathematical model to predict the performance of a random scan in terms of IT and PIR, based on emitter and receiver parameters. Compared to state-of-the-art models, we consider more realistic scenarios, including frequency agile emitters, multichannel random scan, and the addition of switch time between dwells. We provide an explicit and explainable model that covers a wide range of parameter values.

Our model is quantitatively validated using simulation data across multiple configurations and remains consistent with other state-of-the-art models within their respective validity domain, while being more generic. We also demonstrate that our model can be extended to handle emitters with nonperiodic PRIs, and to a certain extent, nonperiodic illumination patterns.

Based on our model, we apply a minimax optimization scheme to derive the optimal dwell time for a given set of emitters to be intercepted. This method is extended, to consider potential estimation errors on the emitter parameters, thus minimizing the risk of missing incorrectly estimated emitters.

Future work could include the implementation in adaptive receivers of algorithms based on the developed model. Because of its robustness to uncertainty in the

emitter parameters, a plausible use for our model and optimization method would be in systems that learn the emitters' parameters based on previous intercepts. The objective would be to estimate the emitters' parameters during the observation phase and use this partial information to improve subsequent intercepts in an online manner.

In such cases, multiple issues with the current model should be addressed. First, the assumption of fully random and uncorrelated scanning strategies between each channel would need to be relaxed. This could be done by considering different mean revisit times for each frequency band. Then, the complexity of the model should be optimized for real-time operation, by using approximations for the integrals and sums when computing the POI. Real-time architectures for model-based strategies could also be used [19].

Adapting the optimization method for real-time adaptation of scanning parameters is also a subject for future work, which can be divided into two parts: 1) the efficient estimation of unknown emitters' parameters using information from past observations; and 2) the real-time strategy for choosing the optimal parameter value at each instant.

APPENDIX

A Computation of the Effective Frequency POI

To compute $\mathbb{P}(\tilde{I}_f|I_t)$, we need to find first $\tilde{n}_{ef}$, the mean number of frequency bands which are effectively active during an observation. Using (4), we know that the mean number of same-frequency pulse groups present during one observation is $n_g = 1 + \tau_d/\tau_g$. Moreover, each pulse group is emitted inside a band of width B_p . The mean number of receiver bands (of width B) intersected by the pulses is $n_p = 1 + B_p/B$. We assume that each pulse group draws n_p random bands with replacement, among the n_{ef} possible ones, with a probability density over bands $\mathbf{p}_{ef}$. For a band b , the probability of being chosen at least once is equal to $1 - (1 - p_{ef_b})^{n_p n_g}$. Thus, the expected value of the number of distinct chosen bands, which corresponds to the effective mean number of active bands $\tilde{n}_{ef}$ is

$$\tilde{n}_{ef} = \sum_{b=1}^{n_{ef}} (1 - (1 - p_{ef_b})^{n_p n_g}). \quad (29)$$

Replacing n_{ef} by $\tilde{n}_{ef}$ in (10) gives $\mathbb{P}(\tilde{I}_f|I_t)$.

ACKNOWLEDGMENT

This work was carried out as part of the MUSHA project. The authors would like to warmly thank the DGA / AID (Directorate General of Armaments / Defense Innovation Agency) for its financial support and for the various high-level scientific exchanges.

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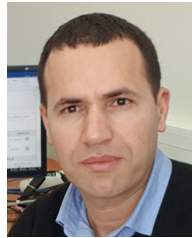
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